WASP-11b

R-1006

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_161130` · created 2026-07-30T02:02:13+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457722.781 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1285 +/- 0.0071
Transit depth (Rp/Rs)^2	1.65 +/- 0.18 [%]
Orbital Inclination (inc)	89.96 +/- 0.46
Airmass coefficient 1 (a1)	1.1097 +/- 0.0077
Airmass coefficient 2 (a2)	-0.096 +/- 0.021
Scatter in the residuals of the lightcurve fit is	0.7 %
Best Comparison Star	None
Optimal Aperture	4.15
Optimal Annulus	5.75
Transit Duration (day)	0.1042 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1285 ± 0.0071 vs 0.131 (deviation 0.35σ)
Duration fit vs published	0.1042 d vs 0.1075 d (deviation 2.36σ)
Mid-time O–C	0.3 min
Depth SNR	18.1
Residual scatter	0.7 %

Light curve

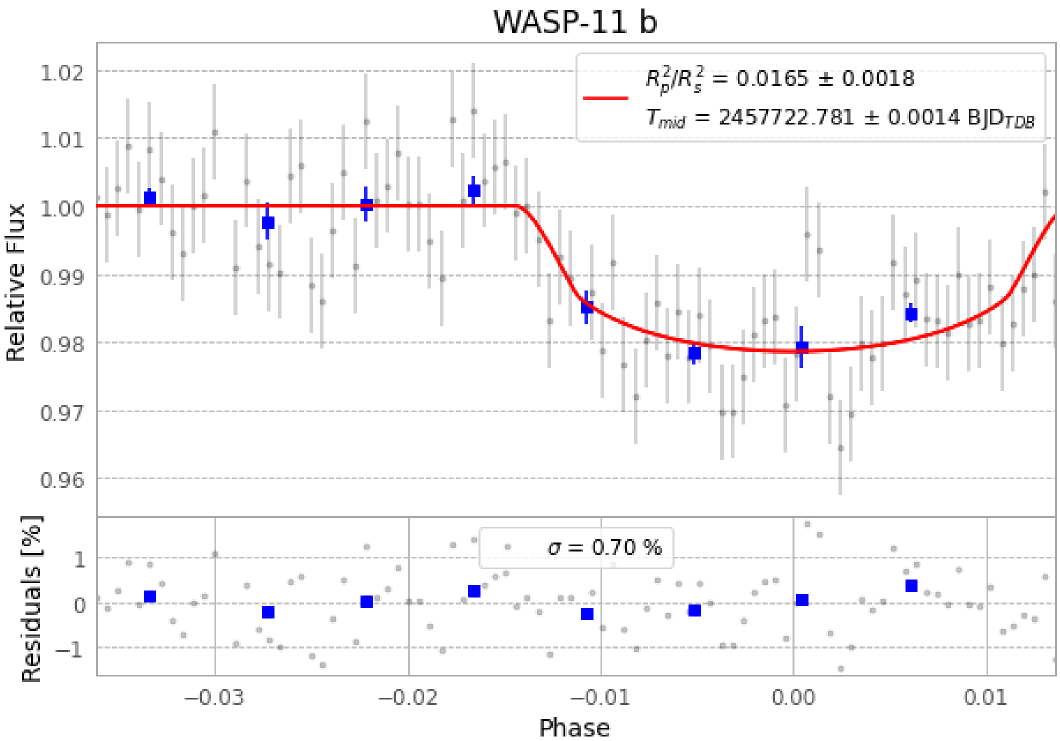

Figure 1 — FinalLightCurve_WASP-11 b_2016-11-29.png (SHA-256 5ce74b7f3dc2a76a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [304, 253], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6caeae251ff696b1…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

90 FITS frames (59 MB), `HATP-10161130032112.FITS` → `HATP-10161130074812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-161130.log (0b9749474a60…), FinalLightCurve_WASP-11 b_2016-11-29.png (5ce74b7f3dc2…), FinalLightCurve_WASP-11 b_2016-11-29.csv (93062bfcc48d…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)